

1. (1) $\Omega \in \mathcal{F}; A \in \mathcal{F} \Rightarrow A^c \in \mathcal{F}; \{A_n\} \in \mathcal{F} \Rightarrow \cup_{n \geq 1} A_n \in \mathcal{F}$
 (2) $P : \mathcal{F} \rightarrow [0, \infty]$ and $P(\emptyset) = 0, P(\Omega) = 1, \{A_n\}_{n \geq 1} \subset \mathcal{F}$ are disjoint $\Rightarrow P(\cup_{n \geq 1} A_n) = \sum_{n \geq 1} P(A_n)$.
 (3) $X : \Omega \rightarrow \mathbb{R}$ and $\forall B \in \mathcal{B}(\mathbb{R}), X^{-1}(B) \in \mathcal{F}$, i.e. X is a real-valued meas. transformation.
 (4) $\exists A \in \mathcal{F}$ s.t $P(A) = 1$ and $\forall \omega \in A, X_n(\omega) \rightarrow X(\omega)$ as $n \rightarrow \infty$, i.e. $P(X_n \rightarrow X) = 1$.
 (5) $P(|X_n - X| > \epsilon) \rightarrow 0$ as $n \rightarrow \infty, \forall \epsilon > 0$.
 (6) $E(|X_n - X|^p) \rightarrow 0$ as $n \rightarrow \infty$.
 (7) $\lim_{t \rightarrow \infty} \sup_{n \geq 1} E(|X_n| I[|X_n| > t]) \equiv \lim_{t \rightarrow \infty} \sup_{n \geq 1} \int_{\{|X_n| > t\}} |X_n| dP = 0$.
 (8) X is a (real valued) random variable with law P_X given by *Exponential*(1) probability on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$, i.e. $P_X(A) = \int_A e^{-u} I_{(0, \infty)}(u) dm(u)$.
 (9) X is a r.v. with law P_X is discrete if $\exists A \in \mathcal{F}$ countable, such that $P_X(A) = 1$.
 (10) If the joint distribution of (X, Y) is the product of the marginal distributions P_X and P_Y , i.e. $P_{(X, Y)} = P_X \times P_Y$.
2. (a) $\exists B \in \mathcal{F}$ such that $\mu_1(B) = \nu_1(B^c) = 0$. Hence, $\mu_0(B) = \nu_0(B^c) = 0$, proving the claim.
 (b) Here X, Y are discrete random variables supported on $\mathbb{N}$. Hence $X + Y$ is discrete, and it is enough to describe $P_{(X+Y)}(\{n\}) = P(X + Y = n) = \sum_{i=0}^{\infty} P(X = i, Y = n - i) = \sum_{i=0}^{\infty} P(X = i)P(Y = n - i) = \sum_{i=0}^{\infty} p_i q_{n-i} = (\tilde{p} * \tilde{q})(n)$ for $n = 0, 1, \dots$.
3. (a) It is easy to show $F_X(x) = (1 - e^{-x})\mathbf{I}_{(0, \infty)}(x)$ - so X is *Exponential*(1) random variable and Y is *Bernoulli*($\frac{2}{3}$) random variable with $F_Y(x) = \frac{1}{3}\mathbf{I}_{(-\infty, x]}(0) + \frac{2}{3}\mathbf{I}_{(-\infty, x]}(1)$. For $x > 0$ $F_Z(x) = P(X^2 \leq x) = P(X \leq \sqrt{x}) = F_X(\sqrt{x})$ and $F_Z(x) = 0$ for $x < 0$.
 (b) Define $\mu^{(1)} = \mu_{F_Z}$ the distribution of Z and $\mu^{(n+1)} = \mu^{(n)} \times \mu$, for $n \geq 1$. For each $n \geq 1$, μ^n is the product measure generated by μ_{F_Z} on $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$ and $\mu^{(n+1)}(A \times \mathbb{R}) = \mu^n(A)$ for $A \in \mathcal{B}(\mathbb{R}^n)$. Hence, from Kolmogorov's Consistency theorem we can construct a sequence of random variables $\{Z_n : n \geq 1\}$ on some probability space $(\Omega, \mathcal{F}, P)$ such that joint distribution of $(Z_1, \dots, Z_n)$ is given by μ^n . Hence $\{Z_n : n \geq 1\}$ is an i.i.d. sequence, each with distr. μ_{F_Z} .
 (c) Using change of variable formula, we can compute $E(Z_n) = E(Z)$ for $n \geq 1$. $E(Z) = \int Z dP = \int X^2 dP = \int x^2 dP_X(x)$ = second moment of *Exponential*(1). So $E(Z) = 2$. Using linearity of integrals, We have $E(T_n) = E(Z_n) = E(Z) = 2$.
 (d) Since X, Y are independent, the distribution of $X + Y$ is $P_X * P_Y$, and

$$\begin{aligned}
 (P_X * P_Y)(A) &= \int \int I_A(x+y) dP_X(x) dP_Y(y) = \int \int I_A(x+y) f_X(x) dm(x) dP_Y(y) \\
 &= \int I_A(x) \left(\int f_X(x-y) dP_Y(y) \right) dm(x) \quad (\text{by Fubini's theorem}) \\
 &= \int_A ((f_X * P_Y)(x)) dm(x)
 \end{aligned}$$

Here $f_X(x) = e^{-x}\mathbf{I}_{(0, \infty)}(x)$ is the *Exponential*(1) density. So $P_X * P_Y \ll m$ and the RN derivative is

$$(f_X * P_Y)(x) = \int f_X(x-y) dP_Y(y) = f_X(x-0)\frac{1}{3} + f_X(x-1)\frac{2}{3}$$